

ACTRIS/EARLINET Level 3 Aerosol Profiling Climatology – Quick Start Guide

Overview

This guide provides a concise set of instructions to help users set up and run the ACTRIS/EARLINET Level 3 aerosol profiling climatology software. The software processes atmospheric lidar data to generate Level 3 NetCDF files with integrated, profile-based, and layer-resolved statistics.

1. Repository and Data

1.1 Clone the Repository

Run the following lines of code from the terminal:

1. `git clone https://github.com/actris-ares/actris-level3-aerosol-profiling-climatology.git`
2. `cd actris-level3-aerosol-profiling-climatology`

1.2 Download Level 2 Data

Download Level 2 NetCDF files for the desired stations from the ACTRIS/EARLINET Data Portal (<https://data.actris-ares.eu>, <https://api.actris-ares.eu/api/swagger-ui/>).

2. Folder Structure

Ensure the following folders exist:

<code>./New/</code>	# Contains Level 2 NetCDF files organized by station
<code>./Layers/</code>	# Contains layer files (one per station)
<code>./Level3/Profiles/</code>	# Output: Level 3 profile NetCDFs
<code>./Level3/Integrated/</code>	# Output: Level 3 integrated NetCDFs
<code>./Level3/Layers/</code>	# Output: Level 3 layer histogram NetCDFs

3. Required Files

Place the following files in the root directory:

- station.csv – Station metadata
- Climatol2.log – Log file for climatological filtering
- Calipso2.log – Log file for CALIPSO filtering

4. Configuration

Before running the main script, open Main.R and configure the following variable:

Release

Set the time range for processing:

1. `release <- c(2000, 2021)` # Example: process data from 2000 to 2021

5. Execution

Run the Main Script

In an R session or terminal:

1. `source("Main.R")`

This will execute the full processing pipeline and generate NetCDF files in the appropriate Level3/ subdirectories.

6. System Requirements

- **Tested OS:** Ubuntu Mate 22.04 (Linux).
- **Recommended OS:** All OS on which the R environment is installed.
- **R Packages Required:**
 - isotone
 - ncdf4
 - radiant.data

- dplyr
- tidyr

No external dependencies beyond R packages are required.

7. Output

The following NetCDF files will be generated:

- Level3/Profiles/<station>/ – Profile-based statistics
- Level3/Integrated/<station>/ – Integrated statistics
- Level3/Layers/<station>/ – Layer-resolved histograms

Each file follows ACTRIS conventions and includes metadata such as time bounds, wavelength, and statistical descriptors.